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Cybersecurity Assignment

Activity 1 : Incident Detection

1. → Multiple failed login attempts.
   → Unusual or abnormal server traffic.
   → Suspicious emails reported by students.

2. → Intrusion Detection System (IDS)
   → Network monitoring tools
   → Antivirus software
   → Email security filters
   → Log monitoring tools.

3. → IT Security team or incident response team.

Activity 2 : Containment Strategy

1 and 2.
- Disconnect infected sys.
  prevents malware from spreading to other computers.
- Block malicious IP address
  stops attackers from continuing communication with the system
- Disable affected user accounts.
  prevents compromised accounts from being misused.

Activity 3 : Eradication

1. → Delete malicious files
   → Run antivirus scans
   → Remove infected email attachments

2. → Phishing emails
   → weak email filtering
   → outdated software or missing security patches

3. → patch vulnerabilities
→ update antivirus & firewal rules.
→ Reset user passwords.
→ Train

why fail without containment?
because malware can continue spreading if infected systems are not isolated first.

Recovery: Activity 4

1. → from clean backups.
→ reinstall affected software

2. → system security patches
→ malware scans
→ network security config.

3. → regularsec updates
→ check ups.
→ strong email filtering.

Activity 5: Digital Forensics Investigation

1. Identifies attacker
Determines how happened
collects legal advices

2. to maintain integrity for legal use

3. · login attempts · user activity · attacker IP address

Chain of custody

1. detection & containment

2. · ignoring warnings
· delayed response
· poor documentation

3. · track events
· supports legal action
· improves future response